

VID (VIRTUAL IDENTIFICATION)

1. PRODUCT VISION & OBJECTIVE

Build VID (Virtual Identification) as a centralized, multi-institution educational management ecosystem.

VID must manage the complete institutional lifecycle:

Admission → Student → Academics → Faculty → Timetable → Attendance → Examinations → Results → Finance → Documents → Staff/HRMS → Parent/Student Engagement → AI Assistance

VID must behave as one connected ecosystem, not as a collection of disconnected CRUD modules.

Core principle

One institution → one connected ecosystem → one student master record → multiple connected services.

The platform must support:

- Multi-institution architecture
- Role-based access
- Modular workspaces
- API-first architecture
- Responsive Web Application
- Parent/Student Mobile Application
- AI-powered services through AI Yantra
- Optional institution-specific modules

2. FINAL WORKSPACE CLASSIFICATION

Core Workspaces

Workspace Interface

- | | | |
|---|-------------------|-----|
| 1 | Super Admin | Web |
| 2 | Institution Admin | Web |
| 3 | Admissions | Web |
| 4 | Academics | Web |
| 5 | Faculty | Web |
| 6 | Attendance | Web |

- 7 Examinations Web
- 8 Finance & Fee Management Web + App
- 9 Documents Web + App
- 10 Staff Management / HRMS Web
- 11 Timetable Web + App
- 12 Yantra Voice Agent Web
- 13 Yantra AI Attendance Web
- 14 Yantra AI Tutor Web + App

Optional Workspaces

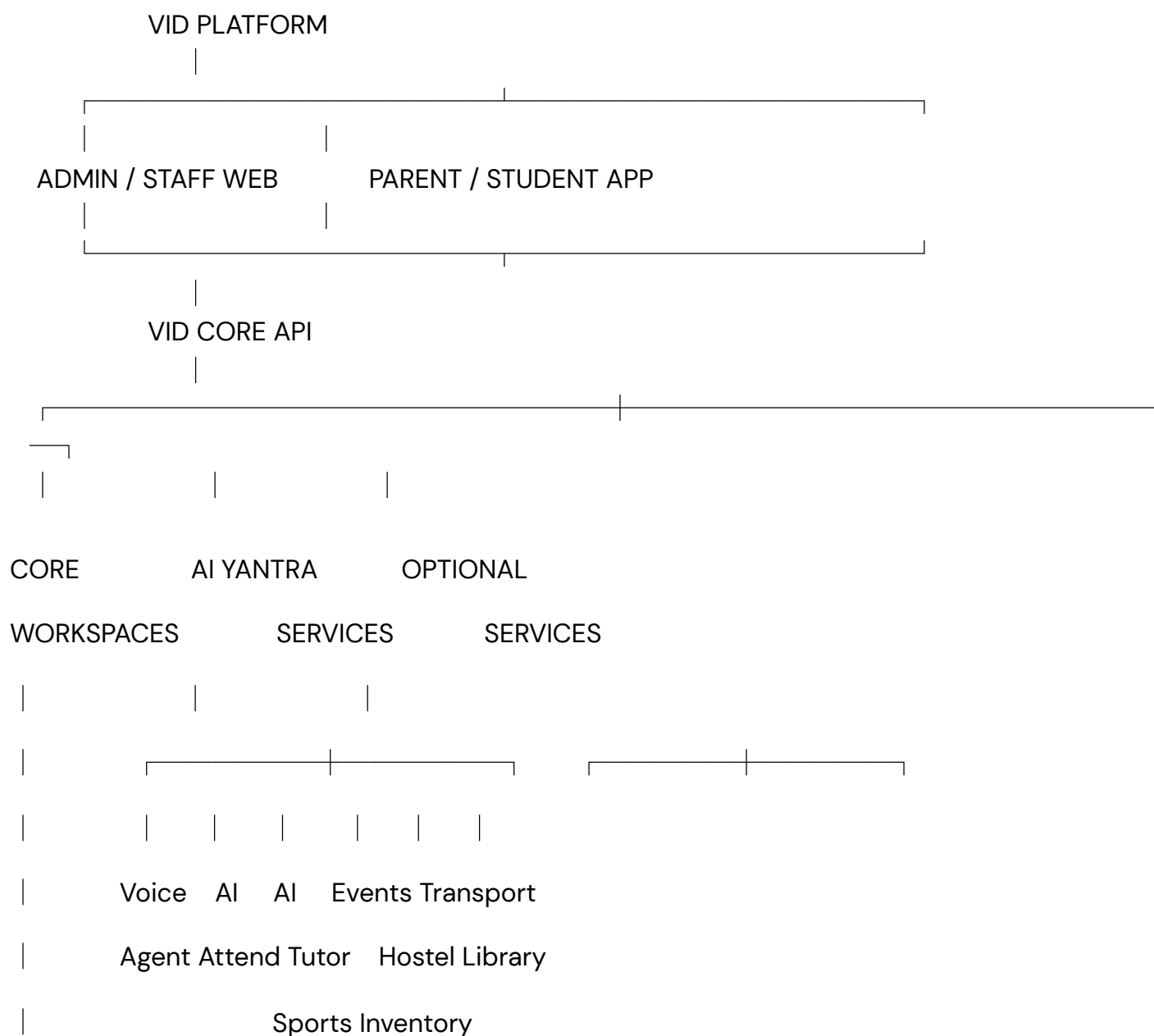
Workspace Interface

- 1 Events Web + App
- 2 Transport Management Web + App
- 3 Hostel Management Web + App
- 4 Library Management Web + App
- 5 Sports Management Web + App
- 6 Inventory & Asset Management Web

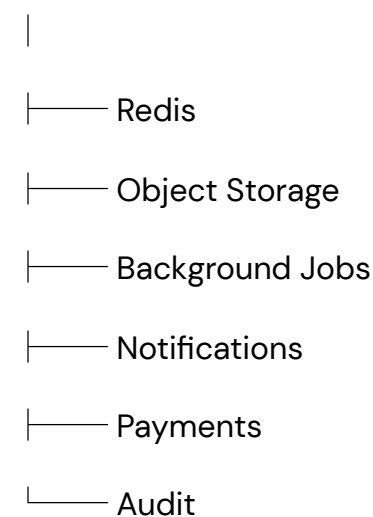
Students are NOT a separate workspace. Students are a central entity used throughout VID.

Optional modules must be independently enabled/disabled per institution.

3. OVERALL PLATFORM ARCHITECTURE



PostgreSQL

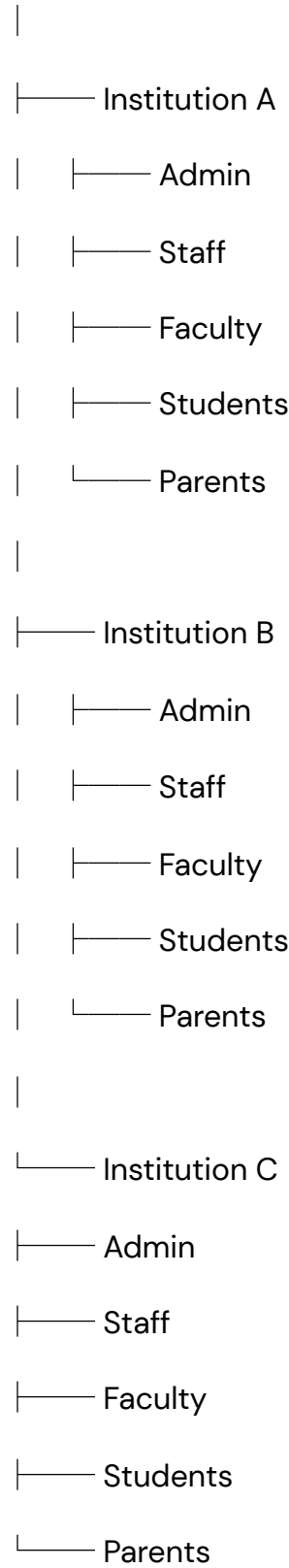


The frontend, mobile application, backend, AI services, database and shared services must communicate through clearly defined interfaces.

4. MULTI-TENANT ARCHITECTURE

VID must support multiple institutions from one SaaS platform.

VID SaaS



Every tenant-owned record must contain:

institution_id

Tenant isolation must be enforced at:

- API level
- Authorization level
- Database query level
- File-storage level
- AI context level

No institution user may access another institution’s data.

5. ROLES, RBAC & AUTHORIZATION

Roles

Role	Access Scope
Super Admin	All institutions
Institution Admin	One institution
Admission Team	Admissions
Academic Coordinator	Academic data
Faculty	Assigned classes/subjects
Exam Team	Examinations
Finance Team	Finance
HR / Staff	HRMS
Student	Own data

Every protected operation must validate:

User

+

Role

+

Institution

+

Permission

+

Resource

+

Action

Example:

Faculty

↓

attendance.create

↓

Class 8-A

↓

Mathematics

↓

Authorized Students

Faculty must not automatically access unrelated classes, finance records or institution-wide data.

6. SUPER ADMIN & INSTITUTION ADMIN

Super Admin

Area	Pages
Dashboard	Dashboard
Institutions	Institutions, Institution Onboarding
Plans	Institution Plans, Subscriptions
Users	Institution Users
Security	Global Roles, Global Permissions
Monitoring	Platform Audit Logs, System Health
AI	AI Yantra Configuration
Billing	Billing
Settings	Platform Settings

Super Admin can:

- Create institutions
- Activate/deactivate institutions
- Create institution administrators
- Manage subscriptions
- Configure global permissions
- Configure global AI services
- Monitor system health
- View platform audit activity

Institution Admin

Must manage:

- Institution Profile
- Academic Years
- Departments
- Courses
- Classes
- Sections
- Subjects
- Users
- Roles

- Permissions
 - Admissions
 - Academics
 - Faculty
 - Attendance
 - Examinations
 - Finance
 - Documents
 - HRMS
 - Timetable
 - AI Yantra
 - Enabled optional modules
-

7. ADMISSIONS & STUDENT LIFECYCLE

Admissions is the starting point of the student lifecycle.

Admissions Pages

Category	Pages
Dashboard	Admissions Dashboard
Enquiries	Enquiries
Applications	Applications, Applicants
Verification	Application Review, Document Verification
Decision	Admission Approval, Rejected, Waitlisted
Registration	Student Registration
Student	Student Profile
Parent	Parent / Guardian Details
Allocation	Class, Section, Academic Year
Identification	Roll Number, Student ID
Documents	Admission Documents
Finance	Admission Fee
History	Admission History

Lifecycle

Enquiry



Application



Document Submission



Verification



Interview / Review



Approval



Student Creation



Parent Creation



Class Allocation



Section Allocation



Roll Number



Student ID



Academic Assignment



Active Student

8. STUDENT MASTER ENTITY

There must be no Student Workspace.

Student is a central master entity.

STUDENT



├── Personal Information

├── Parent / Guardian Information

|—— Admission Information

|—— Academic Information

|—— Attendance

|—— Examinations

|—— Finance

|—— Documents

|—— Timetable

|—— AI Tutor

|—— Optional Services

All related modules must reference the same student record.

This prevents duplicate student records and disconnected information.

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9. **ACADEMICS WORKSPACE**

Pages

Category	Pages
Dashboard	Academic Dashboard
Structure	Academic Years, Departments, Courses, Classes, Sections
Subjects	Subjects
Curriculum	Curriculum, Syllabus
Calendar	Academic Calendar, Academic Sessions
Assignment	Faculty Assignment, Class Assignment, Subject Assignment
Students	Student Academic Records
Progress	Academic Progress
Promotion	Promotion / Demotion
Configuration	Academic Configuration

Academic Hierarchy

Institution



Academic Year



Department



Course



Class



Section



Subjects



Faculty



Students



10. **FACULTY WORKSPACE**

Faculty must have a focused workspace.

Pages

Area	Pages
Dashboard	Faculty Dashboard
Profile	My Profile
Teaching	My Classes, My Sections, My Subjects
Students	My Students, Student Academic Profile
Schedule	Today’s Timetable
Attendance	Attendance
Teaching	Assignments, Study Materials
Assessment	Marks, Examinations
AI	AI Tutor
Performance	Student Performance
HR	Leave, Workload

Faculty permissions must be scoped to assigned:

- Classes
- Sections
- Subjects
- Students
- Timetable

11. ATTENDANCE WORKSPACE

Attendance is a mandatory core workspace.

Attendance Sources

Source	Purpose
AI Face	Yantra AI Attendance
CCTV	Camera-based recognition

Pages

- Attendance Dashboard
- Daily Attendance
- Class Attendance
- Section Attendance
- Subject Attendance
- Faculty Attendance Entry
- Student Attendance
- Attendance History
- Absentees
- Late Arrivals
- Leave Records
- Attendance Corrections
- Attendance Approval
- Attendance Calendar
- Attendance Configuration
- Attendance Logs

Flow

Attendance Source



Recognition / Entry



Validation



Attendance Event



Teacher/Admin Verification



Final Attendance



Parent Notification

AI-generated attendance must pass validation before becoming final attendance.



12. EXAMINATIONS WORKSPACE

Pages

Category	Pages
Dashboard	Exam Dashboard
Configuration	Exam Types, Exam Configuration
Schedule	Exam Schedule, Exam Subjects
Rooms	Exam Rooms
Allocation	Student Allocation, Invigilators
Papers	Question Papers
Marks	Marks Entry, Marks Import
Verification	Marks Verification, Marks Approval
Results	Grades, Results
Reports	Report Cards, Student Result History
Notifications	Exam Notifications
Import	Excel Import

Excel Import

Excel Upload



File Validation



Column Validation



Student Matching



Subject Matching



Marks Validation



Preview



Errors / Warnings



Admin Confirmation



Database Commit

Invalid data must never be silently committed.



13. FINANCE & FEE MANAGEMENT

Finance is a core workspace.

Pages

Category	Pages
Dashboard	Finance Dashboard
Configuration	Fee Structure, Categories, Groups
Assignment	Student Fees, Fee Assignment
Billing	Invoices, Pending Fees
Collection	Fee Collection, Payments
Online	Online Payments, Payment Gateway
Adjustments	Discounts, Scholarships
Refunds	Refunds
Receipts	Receipts
Reconciliation	Payment Reconciliation
Reports	Financial Reports
Audit	Finance Audit Logs

Payment Flow

Fee Assignment



Invoice



Parent Payment



Payment Gateway



Gateway Response



Verification



Transaction



Receipt



Fee Status Updated

Transactions must be auditable and historical records must not simply be overwritten.



14. DOCUMENTS & HRMS

Documents

Pages

- Document Dashboard
- Student Documents
- Staff Documents
- Institution Documents
- Admission Documents
- Certificates
- Bonafide Certificates
- ID Documents
- Circulars
- Templates
- Document Verification
- Document Requests
- Document History

Supported:

PDF, JPG, PNG, DOC, DOCX

HRMS

Category	Pages
Dashboard	HRMS Dashboard
Staff	Staff Directory, Staff Profiles
Organization	Departments, Designations
Employment	Employment Details
Documents	Staff Documents
Attendance	Staff Attendance
Leave	Leave Management, Requests, Approval
Workload	Workload, Faculty Workload
History	Staff History
Payroll	Payroll Integration
Reports	HR Reports

Payroll can initially remain an external integration boundary.

15. TIMETABLE WORKSPACE

Timetable is a core workspace.

Pages

- Timetable Dashboard
- Academic Timetable
- Class Timetable
- Section Timetable
- Faculty Timetable
- Subject Timetable
- Room Timetable
- Period Configuration
- Working Days
- Breaks

- Substitution
- Free Periods
- Timetable Conflicts
- Timetable Publishing

Conflict Detection

The system must detect:

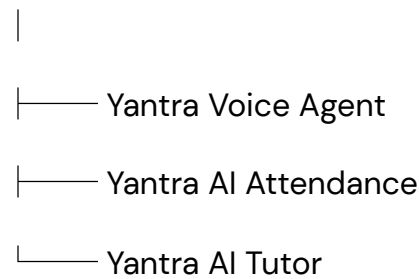
Conflict	Example
Teacher	Teacher assigned twice
Room	Room assigned twice
Class	Class receives conflicting subjects
Section	Section receives conflicting periods
Period	Invalid period allocation

16. AI YANTRA

AI Yantra is the intelligence layer of VID.

It contains exactly three workspaces:

AI YANTRA



Workspace	Purpose	Interface
Voice Agent	AI communication	Web
AI Attendance	Face recognition attendance	Web
AI Tutor	Personalized learning	Web + App

Do not create additional AI Yantra workspaces unless the product definition is explicitly changed.

17. YANTRA VOICE AGENT

Pages

Area	Pages
Dashboard	Voice Agent Dashboard
Campaigns	Campaigns, Create Campaign
Recipients	Call Recipients
Templates	Call Templates
AI	Conversation Configuration
Languages	Languages
Voice	Voice Configuration
Queue	Call Queue
Calls	Active, Completed, Failed
History	Call History, Interaction History
Scheduling	Scheduled Calls
Logs	Voice Agent Logs

Use Cases

- Fee Reminder
- Attendance Alert
- Exam Reminder
- PTM Reminder
- Event Announcement
- Admission Follow-up
- Emergency Announcement
- General Announcement

Campaign



Audience



Language



AI Voice Agent



Call



Conversation



Response



Interaction Record



18. YANTRA AI ATTENDANCE

Pages

- AI Attendance Dashboard
- Face Registration
- Student Face Profiles
- Face Recognition

- Camera Management
- CCTV Devices
- Classroom Cameras
- Live Recognition
- Attendance Verification
- Recognition Logs
- Unknown Faces
- Attendance Corrections
- Device Health
- AI Attendance Configuration

Flow

Face Registration



Face Profile



Face Embedding



Encrypted Storage



Camera Stream



Face Detection



Face Recognition



Confidence Check



Student Identification



Attendance Event



Attendance Validation



Core Attendance

Biometric representations must use strong security controls, access restrictions and retention policies.



19. YANTRA AI TUTOR

Pages

Area	Pages
Dashboard	AI Tutor Dashboard
Subjects	My Subjects
Learning	Ask Tutor, Topic Explanation, Doubt Resolution
Practice	Practice Questions, Question Generator
Exams	Mock Exams
Analysis	Weak Topic Detection, Performance Analysis
Recommendations	Learning Recommendations, Personal Study Plan
History	Learning History
Progress	Progress Tracking

Flow

Student



Authorized Academic Data



Performance Data



Weak Topic Detection



AI Tutor



Explanation



Practice



Assessment



Recommendation



Learning Plan

AI Tutor must only access authorized student/context data.



20. OPTIONAL MODULES

Optional modules:

- Events
- Transport
- Hostel
- Library
- Sports
- Inventory & Assets

They must be:

1. Independently enabled/disabled.
2. Hidden from navigation when disabled.
3. Isolated from core dependencies.

- 4. Connected to shared Student/User entities where required.
- 5. Accessible only according to permissions.

Module Summary

Module	Main Capabilities
Events	Events, registrations, participants, attendance, certificates
Transport	Routes, stops, vehicles, drivers, tracking, allocation
Hostel	Hostels, rooms, beds, allocation, attendance, fees
Library	Books, copies, issue, return, fines, reservations
Sports	Teams, coaches, training, competitions, results
Inventory	Assets, stock, vendors, purchases, allocation, maintenance

21. PARENT & STUDENT MOBILE APPLICATION

Use one mobile application with role-based experiences.

MOBILE APP

|

|_____ STUDENT

|_____ PARENT

Student Bottom Navigation

Navigation	Content
Home	Overview
Academics	Classes, subjects, materials
Attendance	Attendance
Fees	Finance
Profile	Personal information

Student Home should display:

- Welcome
- Today's Timetable
- Today's Attendance
- Upcoming Exams
- Pending Fees
- Assignments
- AI Tutor
- Announcements

Additional sections:

- Academics
- Attendance
- Examinations
- Finance
- AI Tutor
- Documents
- Timetable

Parent Experience

Parents must be able to manage multiple children from one login.

Parent

└── Child 1

└── Child 2

└── Child 3

Parent features:

- Child Profile
- Attendance
- Academics
- Exams
- Fees
- Timetable
- Documents
- Notifications
- AI Tutor where permitted

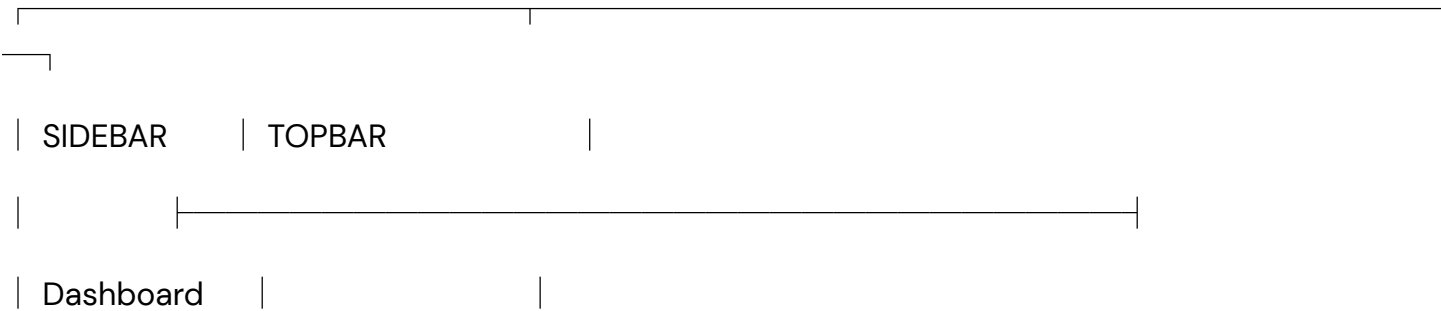
22. RESPONSIVE WEB DESIGN

The entire admin/staff web application must be fully responsive.

Supported Widths

Device	Width
Small Mobile	320px+
Mobile	375px+
Large Mobile	390px+
Large Mobile	414px+
Tablet	768px+
Small Desktop	1024px+
Desktop	1280px+
Large Desktop	1440px+

Desktop



23. RESPONSIVE COMPONENT BEHAVIOR

Tables

Desktop:

- Sorting
- Filtering
- Pagination
- Bulk actions
- Column visibility

Mobile:

- Convert complex rows into cards where appropriate.
- Use internal table scrolling when a true table is necessary.
- Never allow page-level horizontal scrolling.

Forms

Width	Layout
320–414px	1 column
768px	1–2 columns
1024px	2 columns
1280px+	2–4 columns

Forms must support:

- Inline validation
- Responsive fields
- Searchable selectors
- Mobile date pickers
- Bottom sheets
- Sticky actions where appropriate
- Clear error states

Dashboards

Cards and charts must automatically reflow.

Desktop:

4 cards → 2 charts

Mobile:

1 card

1 card

1 card

1 card

1 chart

1 chart

24. TECHNICAL STACK & BACKEND ARCHITECTURE

Recommended Stack

Layer	Technology
Frontend	React / Next.js
Language	TypeScript
Styling	Tailwind CSS
Data Fetching	TanStack Query
Validation	Zod
Backend	Python + FastAPI
Database	PostgreSQL
Cache	Redis
Jobs	Celery + Redis / Managed Queue
Storage	S3-compatible object storage
AI	Python AI services
Vision	OpenCV + Face Recognition Model
LLM	LLM API
Speech	STT + TTS
Vector Search	Vector DB where required

Vector Search Vector DB where required

If the project standardizes on Flask, Flask REST APIs may be used, but FastAPI is preferred for the API-first architecture.

Backend Structure

backend/

├── auth/

- |—— tenants/
- |—— users/
- |—— admissions/
- |—— students/
- |—— academics/
- |—— faculty/
- |—— attendance/
- |—— examinations/
- |—— finance/
- |—— documents/
- |—— hrms/
- |—— timetable/
- |—— events/
- |—— transport/
- |—— hostel/
- |—— library/
- |—— sports/
- |—— inventory/
- |—— ai_yantra/
 - | |—— voice_agent/
 - | |—— ai_attendance/
 - | |—— ai_tutor/
- |—— notifications/
- |—— audit/
- |—— common/

Main Database Domains

Domain	Entities
Platform	institutions
Auth	users, roles, permissions, user_roles
Admissions	admissions, applications, admission_documents
Students	students, parents, guardians, student_parents
Academics	academic_years, departments, courses, classes, sections, subjects
Curriculum	curriculum, syllabus
Faculty	faculty, faculty_subjects, faculty_classes
Attendance	attendance, attendance_records, attendance_sessions, attendance_corrections
Exams	exams, exam_types, exam_schedules, exam_subjects, marks, grades, report_cards
Finance	fee_structures, student_fees, invoices, payments, transactions, receipts, discounts, scholarships, refunds
Documents	documents, document_templates
HRMS	staff, departments, designations, staff_attendance, leave_requests
Timetable	timetables, periods, rooms
Voice	ai_voice_campaigns, ai_voice_calls
AI Attendance	face_profiles, face_embeddings, ai_attendance_events
AI Tutor	tutor_sessions, tutor_messages, learning_profiles, recommendations

Every major tenant-owned entity must include institution_id either directly or through a securely enforced parent relationship.

API Structure

/api/v1/auth

/api/v1/institutions

/api/v1/users

/api/v1/admissions

/api/v1/students

/api/v1/academics

/api/v1/faculty

/api/v1/attendance

/api/v1/examinations

/api/v1/finance

/api/v1/documents

/api/v1/hrms

/api/v1/timetable

/api/v1/events

/api/v1/transport

/api/v1/hostel

/api/v1/library

/api/v1/sports

/api/v1/inventory

/api/v1/yantra/voice

/api/v1/yantra/attendance

/api/v1/yantra/tutor

26. SECURITY, AUDIT, FILES & NOTIFICATIONS

Security

VID must implement:

- HTTPS
- Secure password hashing
- JWT or secure sessions
- Refresh token rotation
- RBAC
- Tenant isolation
- Permission middleware
- Resource-level authorization
- Rate limiting
- Input validation
- Secure headers
- CSRF protection where applicable
- File validation
- Malware scanning
- Private object storage
- Signed URLs

Audit

Critical actions must record:

Field	Example
Actor	Teacher A
Action	Attendance Updated
Resource	Student 1023
Old Value	ABSENT
New Value	PRESENT
Timestamp	Date/time
Institution	Tenant
IP/Device	Where appropriate

Notifications

There is no Communication workspace.

Notifications are shared services.

Source	Recipient
Attendance	Parent/Student
Finance	Parent/Student
Exams	Student/Parent
Admissions	Applicant/Parent
Events	Participants

Channels:

- Push
- SMS
- Email
- Yantra Voice

Files

Use object storage:

institutions/

students/

photos/

documents/

staff/

examinations/

question-papers/

report-cards/

finance/

receipts/

ai/

Database stores metadata and file references, not large binary files.

27. COMPLETE SYSTEM JOURNEYS

Student

Enquiry



Application



Verification



Approval



Student



Parent



Class



Section



Subjects



Faculty



Timetable



Attendance



Exams



Results



Fees



Documents



AI Tutor

Faculty

Login



Dashboard



Today's Timetable



Class



Attendance



Teaching



Assignments / Materials



Marks



Student Performance



AI Tutor Support

Parent

Login



Select Child



Dashboard

Attendance

Academics

Exams

Fees

Timetable

Documents

AI Tutor

Finance

Fee Structure



Fee Assignment



Invoice



Parent



Payment



Gateway



Verification



Transaction



Receipt



Notification



28. MODULE DEPENDENCIES & NAVIGATION

Main Institution Admin Navigation

Dashboard

CORE

|—— Admissions

|—— Academics

|—— Faculty

|—— Attendance

|—— Examinations

|—— Finance & Fees

|—— Documents

|—— Staff / HRMS

|—— Timetable

AI YANTRA

|—— Voice Agent

|—— AI Attendance

|—— AI Tutor

OPTIONAL

|—— Events

- Transport
- Hostel
- Library
- Sports
- Inventory & Assets

Core Dependency

Institution



Admissions



Student

- Academics
- Finance
- Documents
- Faculty



Timetable



Attendance



Examinations



AI Tutor

Optional modules connect to the Student and institution but must never become mandatory dependencies of the core academic engine.

Performance Targets

Metric	Target
Normal page load	< 2.5 seconds
Standard API response	< 500 ms
Standard DB query	< 200 ms
Production availability	99.9%+ where practical

AI operations may naturally require longer processing times.

Development Phases

Phase	Scope
Phase 1	Super Admin, Institution Admin, Auth, RBAC, Admissions, Students, Academics, Faculty
Phase 2	Attendance, Examinations, Finance, Documents, Timetable, HRMS
Phase 3	Yantra Voice Agent, AI Attendance, AI Tutor
Phase 4	Events, Transport, Hostel, Library, Sports, Inventory

Quality Requirements

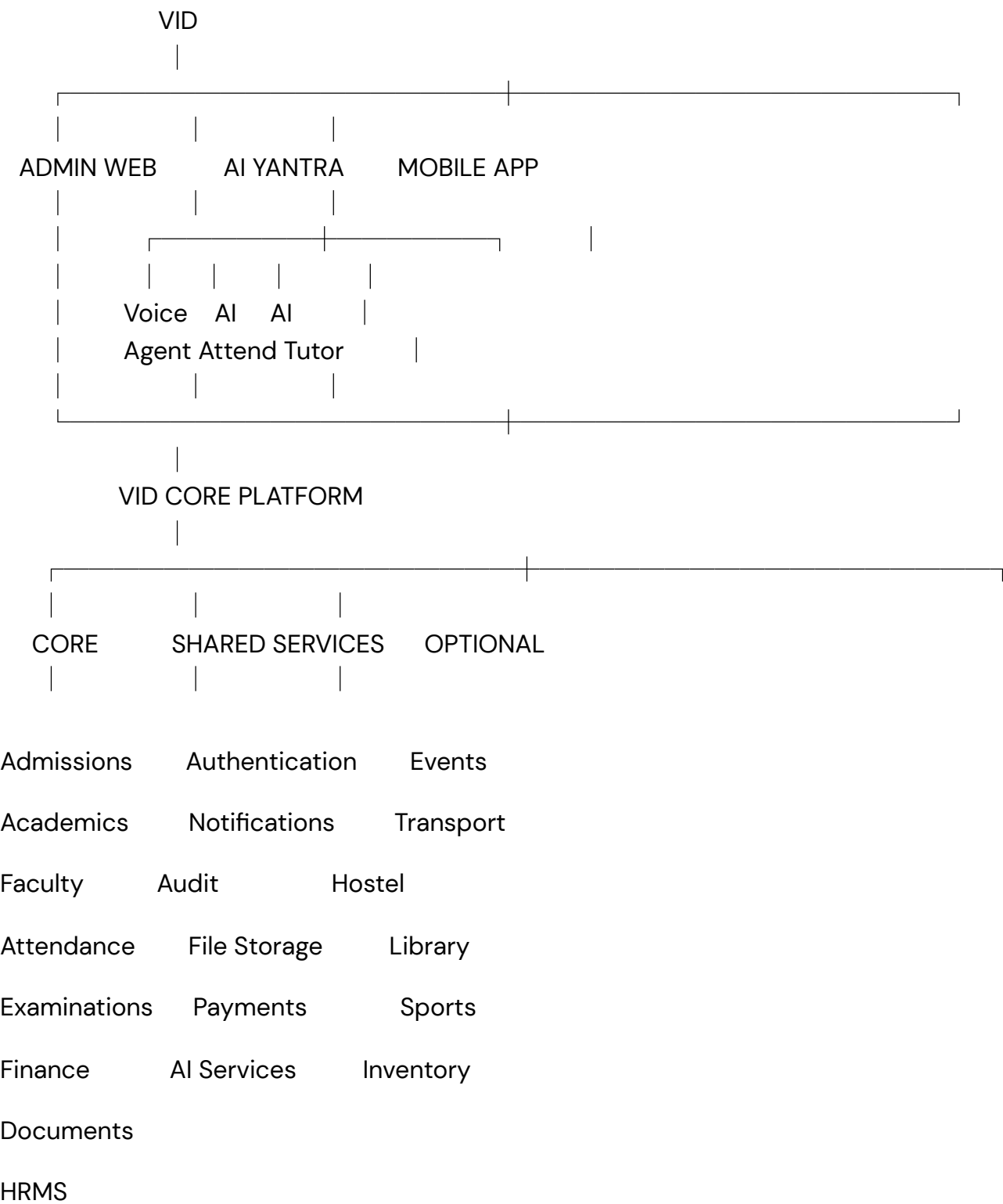
The application must maintain:

- Consistent UI components
- Consistent design system
- Loading states
- Empty states
- Error states
- Success states
- Permission states
- Responsive layouts
- Accessibility
- Form validation
- Search and filtering
- Pagination
- Confirmation for destructive actions

- Proper audit trails
- No broken navigation
- No page-level horizontal scrolling
- No inaccessible optional modules
- No cross-tenant data exposure

30. FINAL IMPLEMENTATION RULES & PRODUCT BOUNDARY

Build VID as one unified educational operating platform.



Timetable

Non-Negotiable Rules

#	Rule
1	Student remains a central master entity, not a separate workspace.
2	Core workspaces are always available.
3	AI Yantra contains exactly three workspaces.
4	Optional modules can be independently enabled or disabled.
5	Disabled modules must disappear from navigation.
6	Every tenant-owned record must be institution-isolated.
7	RBAC must be combined with resource-level authorization.
8	Faculty access must be limited to assigned resources.
9	Parent access must be limited to their children.
10	Students can only access their own permitted data.
11	AI services must never receive unrestricted database access.
12	AI Attendance must pass through validation before final attendance.
13	Financial transactions must remain auditable.
14	Important changes must generate audit events.
15	Large files must use object storage.
16	Notifications remain a shared service, not a workspace.
17	APIs must be versioned.
18	The platform must be API-first.
19	The admin/staff application must be fully responsive.
20	The mobile app must provide dedicated Parent/Student experiences.

21	No page-level horizontal scrolling is allowed.
22	Tables must have responsive mobile behavior.
23	Forms must adapt to screen size.
24	Dashboards must automatically reflow.
25	Navigation must adapt by role and screen size.
26	Security must be enforced server-side, not only through UI hiding.
27	Optional services must never break the core platform.
28	All connected workflows must use shared master entities.
29	Performance, accessibility, security and responsive behavior are product requirements, not later enhancements.
30	The final system must operate as one connected VID ecosystem rather than multiple disconnected applications.

FINAL BUILD INSTRUCTION

Implement the complete VID platform according to these 30 points. Preserve the defined workspace boundaries, role permissions, multi-tenant architecture, connected student lifecycle, AI Yantra structure, optional-module architecture, responsive behavior, API architecture, security requirements, database relationships and user journeys. Do not introduce additional workspaces, unnecessary duplicate entities, disconnected module-specific student records, or desktop-only interfaces unless explicitly required by a future product change.